Week 2: Foundations of Market Microstructure

From limit order books to the role of the market maker

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Learning Objectives

- Reconstruct a limit order book from raw exchange events
- Decompose the bid-ask spread into its three classical components
- Explain the market maker as a continuous liquidity provider

Lecture 1: Order Book Mechanics

- Order types: market, limit, IOC, post-only, hidden
- Price-time priority and the matching algorithm
- L1 / L2 / L3 views; tick size and lot size
- Coincall architecture: contracts, fees, market-maker tiers

Lecture 2: Spread Decomposition

- Adverse selection, inventory, and order-processing costs
- Roll (1984): effective spread from trade prices
- Glosten-Milgrom (1985): asymmetric information
- Kyle (1985): the strategic informed trader
- Realized spread, effective spread, price impact

Recitation

- Build an L2 order-book reconstructor in Python
- Replay one hour of Coincall BTC-PERP events
- Emit a CSV of mid-price, spread, and depth
- Watch out: delete price levels that empty after a cancel

Reading & Problem Set 2 (Python)

- Cartea-Jaikumar-Penalva Ch. 1-3; Roll (1984); Glosten-Milgrom (1985)
- PS2: order book, Roll estimator, Glosten-Milgrom, toxicity